
Lecture Notes

on Baye's Theorem

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1 Bayes' Theorem

Let A and B be events in a sample space S with $P(A) \neq 0$. Then

$$P(B | A) = \frac{P(A | B) P(B)}{P(A)}$$

This result is called **Bayes' Theorem**.

$$P(A|B_i)$$

$$P(B_i|A) \checkmark$$

1.1 Proof

Recall that, for events A and B with $P(B) \neq 0$,

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \Rightarrow P(A \cap B) = P(A | B) P(B). \quad (1)$$

Also, for $P(A) \neq 0$,

$$P(B | A) = \frac{P(B \cap A)}{P(A)} = \frac{P(A \cap B)}{P(A)}.$$

$$(A \cap B) = (B \cap A) \\ \Rightarrow P(A \cap B) = P(B \cap A) \quad (2)$$

Using (1) in (2), we obtain

$$P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A | B) P(B)}{P(A)}.$$

Thus Bayes' Theorem holds.

1.2 Bayes' Theorem with a Partition

Let S be a sample space partitioned by mutually exclusive events

$$B_1, B_2, \dots, B_n,$$

so that $\{B_1, \dots, B_n\}$ is a partition of S and $P(B_i) > 0$ for all i . Let A be an event with $P(A) \neq 0$. Then, for each $k \in \{1, 2, \dots, n\}$,

$$P(B_k | A) = \frac{P(A | B_k) P(B_k)}{P(A)}.$$

Using the Partition (Total Probability) Theorem,

$$P(A) = \sum_{i=1}^n P(A | B_i) P(B_i),$$

so we can write

$$P(B_k | A) = \frac{P(A | B_k) P(B_k)}{\sum_{i=1}^n P(A | B_i) P(B_i)}$$

This is Bayes' Theorem in the presence of a partition of the sample space.

Example. Bag 1 contains 4 white and 6 black balls, and bag 2 contains 4 white and 3 black balls. One ball is drawn at random from one of the two bags and is found to be black. Find the probability that this ball was drawn from bag 1.

Let B_1 = denotes the event of choosing bag 1.
 B_2 = " " " " " " 2.
 A = " " " of drawing a black ball.

$$P(B_1 | A) = ?$$

$$n_1 = n.$$

Given, $P(B_1) = \frac{1}{2} = P(B_2)$

$$P(A | B_1) = \frac{{}^6C_1}{{}^{10}C_1} = \frac{6}{10} = \frac{3}{5}$$

$$P(A | B_2) = \frac{{}^3C_1}{{}^7C_1} = \frac{3}{7}$$

By using Baye's theorem,

$$P(B_1 | A) = \frac{P(A | B_1) P(B_1)}{P(A | B_1) P(B_1) + P(A | B_2) P(B_2)}$$

$$P(A)$$

$$= P(A | B_1) P(B_1)$$

$$+ P(A | B_2) P(B_2)$$

$$= \frac{(3/5) \times (1/2)}{(3/5 \times 1/2) + (3/7 \times 1/2)}$$

$$= \frac{(3/10)}{(3/10) + (3/14)} = \frac{(3/10)}{(21+15)/70} = \frac{(3/10)}{(36/70)}$$

$$= \frac{3}{10} \times \frac{70}{36} = \frac{7}{12}$$

✎ **Example.** A certain disease affects 1% of the population. A diagnostic test for the disease has the following characteristics:

- If a person has the disease, the test is positive with probability 0.95 (sensitivity). $\rightarrow P(A|D) = \frac{TP}{TP+FN}$
- If a person does not have the disease, the test is negative with probability 0.90 (specificity). $\rightarrow P(\bar{A}|\bar{D}) = \frac{TN}{TN+FP}$

A randomly selected person from the population tests positive. What is the probability that this person actually has the disease?

Let D = denotes the event of having the disease.

$\bar{D}$ = " " " " not " " "
 A = " " " " the test report to be positive.
 $\bar{A}$ = " " " " " " " " negative.

Q: $P(D|A) = ?$

Given, $P(D) = 0.01$, $P(\bar{D}) = 1 - P(D) = 1 - 0.01 = 0.99$.

$P(A|D) = 0.95$

$P(A|\bar{D}) = 1 - P(\bar{A}|\bar{D}) = 1 - 0.90 = 0.10$.

$P(A|\bar{D}) = 1 - P(\bar{A}|\bar{D})$ Thus by the Bayes' theorem,

$$P(D|A) = \frac{P(A|D) P(D)}{P(A|D) P(D) + P(A|\bar{D}) P(\bar{D})}$$

$$= \frac{(0.95 \times 0.01)}{(0.95 \times 0.01) + (0.10 \times 0.99)}$$

$\approx 0.09 = 9\%$.